

THIS IS A TEST FILE

Package **intexgral** Version: 3.0.1 - 2026-01-02

A LaTeX package for typesetting integrals.

Simple test

$$\int x \, dx \quad (1)$$

Key **limits** test

$$\int_a^b x \, dx \quad (2)$$

$$\int_a^b \int_c^d \int_e^f x \, dx \quad (3)$$

Key **limits*** test

$$\int_{[a,b]} x \, dx \quad (4)$$

$$\int_{[a,b]} \int_{[c,d]} \int_{[e,f]} x \, dx \quad (5)$$

Key **mode** test

$$\int_a^b \int_c^d \int_e^f f^2 g^3 h^4 \, df \, dg \, dh \quad (6)$$

$$\int_a^b f^2 \int_c^d g^3 \int_e^f h^4 \, df \, dg \, dh \quad (7)$$

$$\int_a^b f^2 \, df \int_c^d g^3 \, dg \int_e^f h^4 \, dh \quad (8)$$

Key **symbol** test

$$\iint x \, dx \quad (9)$$

$$\int x \, dx \quad (10)$$

Key **nint** test

$$\iiint x \, dx \quad (11)$$

Keys `llimit` and `ulimit` test

$$\int_X x \, dx \quad (12)$$

$$\int^X x \, dx \quad (13)$$

Key `domain(*)` test

$$\int_{\mathbb{R} \times \mathbb{Z}} x \, dx \quad (14)$$

$$\int_{\mathbb{R}^2 \times \mathbb{Z}^5 \times \mathbb{C}^9} x \, dx \quad (15)$$

$$\int_{\mathbb{R} \times \mathbb{Z}_p} x \, dx \quad (16)$$

$$\int_{\mathbb{E}_i \times \mathbb{A}_t} x \, dx \quad (17)$$

Key `boundary` test

$$\int_{\partial S} x \, dx \quad (18)$$

Key `variables` test

$$\int_a^b \int_a^b \int_a^b xyz \, dx \, dy \, dz \quad (19)$$

$$\int x \quad (20)$$

Key `jacobian` test

$$\int_a^b \int_a^b r^3 \sqrt{\theta} r \, dr \, d\theta \quad (21)$$

$$\int_a^b \int_a^b \int_a^b r^3 \sqrt{\theta} e^\phi r^2 \sin \theta \, dr \, d\theta \, d\phi \quad (22)$$

Key `diff-symb` test

$$\int x \, \mathcal{D}q(t) \quad (23)$$

Key `diff-star` test

$$\int x \, \mathrm{d}_x \, \mathrm{d}_y \quad (24)$$

Key `diff-options` test

$$\int x \, \mathrm{d}^4 x \quad (25)$$

$$\int_a^b r \, \mathrm{d}^2 (r) \int_a^b \theta \, \mathrm{d}^4 \theta \int_a^b \phi \, \mathrm{d}^6 (\phi) \quad (26)$$

Macro `\NewLimitsKeyword` test

$$\int_{\text{lower}}^{\text{upper}} x \, \mathrm{d}x \quad (27)$$

Macro `\NewVariableKeyword` test

$$\int_a^b \int_a^b \int_a^b x \, \mathrm{d}\alpha \, \mathrm{d}\beta \, \mathrm{d}\gamma \quad (28)$$

Macro `\NewSymbolKeyword`

$$\oint_S x \, \mathrm{d}x \quad (29)$$

Macro `\differentials` test

$$\int \frac{\mathrm{d}x}{x} \quad (30)$$

Special syntax test

$$\int_1^{10} x \, \mathrm{d}x \quad (31)$$

$$\int_a^b r r^2 \, \mathrm{d}r \int_a^b \theta \sin \theta \, \mathrm{d}\theta \int_a^b \phi \, \mathrm{d}\phi \quad (32)$$

$$\int_a^b r^2 \, \mathrm{d}r \int_a^b \sin \theta \, \mathrm{d}\theta \int_a^b \mathrm{d}\phi \quad (33)$$

$$\int_a^b r^2 \, \mathrm{d}r \int_a^b \sin \theta \, \mathrm{d}\theta \int_a^b \mathrm{d}\phi \quad (34)$$

Messages test

$$\int_a^b x \, dx \int_a^b dy \tag{35}$$

$$\int_a^b \int_a^b x \, dx \, dy \, dz \tag{36}$$

$$\int_a^b \frac{1}{x} \int_a^b \frac{dx \, dy}{y} \tag{37}$$

$$\tag{38}$$